COMPARING MEASURES OF EARNINGS INSTABILITY BASED ON SURVEY AND ADMINISTRATIVE REPORTS

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Abstract

In Celik, Juhn, McCue, and Thompson (2009), we found that estimated levels of earnings instability based on data from the Current Population Survey (CPS) and the Survey of Income and Program Participation (SIPP) were reasonably close to each other and to others' estimates from the Panel Study of Income Dynamics (PSID), but estimates from unemployment insurance (UI) earnings were much larger. Given that the UI data are from administrative records which are often posited to be more accurate than survey reports, this raises concerns that measures based on survey data understate true earnings instability. To address this, we use links between survey samples from the SIPP and UI earnings records in the LEHD database to identify sources of differences in work history and earnings information. Substantial work has been done comparing earnings levels from administrative records to those collected in the SIPP and CPS, but our understanding of earnings instability would benefit from further examination of differences across sources in the properties of changes in earnings. We first compare characteristics of the overall and matched samples to address issues of selection in the matching process. We then compare earnings levels and jobs in the SIPP and LEHD data to identify differences between them. Finally we begin to examine how such differences affect estimates of earnings instability. Our preliminary findings suggest that differences in earnings changes for those in the lower tail of the earnings distribution account for much of the difference in instability estimates.

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1 Introduction

The U.S. labor market witnessed an extraordinary increase in wage and earnings inequality during the 1970s and the 1980s. This rise in inequality was comprised of both a growing gap in earnings across individuals as well as growing within-person variance in earnings. Gottschalk and Moffitt (1994) were the first to document the rise in this latter component, often referred to in the literature as earnings instability. They and other papers that followed (for example Haider (2001) and Cameron and Tracy (1998)) roughly reached the following conclusion about earnings instability trends through the mid-1990s: earnings instability increased dramatically during the 1970s and reached a peak during the 1982 recession but after that peak declined to the late 1970s level.

A number of recent papers have examined the question of what has happened to earnings instability since then, including Moffitt and Gottschalk (2008), Gottschalk, McEntarfer, and Moffitt (2008), Dahl, DeLeire, and Schwabish (2008), and Shin and Solon (2008). This work has drawn on a wide variety of data sources, making use of both survey and administrative sources of earnings data. In our previous work Celik, Juhn, McCue, and Thompson (2009), we used three panel datasets (the matched March CPS, the SIPP, and UI earnings records incorporated into the Longitudinal Employer-Household Dynamics (LEHD) database) to examine trends in male earnings instability in recent decades. While the trends in our three data sources appear similar during periods of overlap, estimated levels of instability differ substantially, as Figure 1 illustrates. Levels estimated from the CPS and SIPP are reasonably close to each other (and to others' estimates from the PSID), but the UI estimates are much larger.[1]

Given that administrative records are often posited to be more accurate than survey reports, this raises the question of whether measures based on survey data understate

[1]See Shin and Solon (2008) for PSID estimates based on the same methodology used in Figure 1. Gottschalk, McEntarfer, and Moffitt (2008) use a methodology that differs from ours, but they also note a difference in levels when applying it to UI and PSID earnings.

true earnings instability. To address this question, we use links between UI earnings records and survey samples from the SIPP to identify differences in work history and earnings information and to examine how such differences affect measured earnings instability. Our goal is a better understanding of the relative strengths of alternative data sources in studying earnings instability, but we hope the work will be useful in understanding measures of earnings dynamics more generally.

We carry out three sorts of exercises: We first compare the overall and matched samples to address issues of selection in the matching process. We then compare earnings levels and jobs in the SIPP and LEHD data to identify differences between them. Finally we begin to examine differences in estimates of earnings instability. Our preliminary findings suggest that differences in earnings changes for those in the lower tail of the earnings distribution account for much of the difference in instability estimates.

2 Background

A number of studies have compared the properties of matched survey and administrative data. In this preliminary draft, we briefly mention closely related studies that also use links between Census household surveys and administrative earnings records. While none directly addresses how differences in source data affect measures of earnings instability, they have produced a number of insights which we draw on in this work. Abraham, Haltiwanger, Sandusky, and Spletzer (2009) use UI data matched to CPS respondents. Their focus is on differences in the measures of employment and jobs in the two data sources. They find large discrepancies in employment status, and present evidence that these discrepancies tend to be associated with two patterns: marginal labor force attachment—those with short low-paid jobs in UI data seem to under-report employment in CPS, and non-traditional employment relationships—those with em-

ployment in CPS, but not in the UI data are over-represented in sectors where we might expect to see more non-employee or off-the-books work.

Abowd and Stinson (2005) compare SIPP measures of earnings to annual social security earnings records. They decompose earnings variation into common versus idiosyncratic (to the source) components of earnings. They find that the shared component accounts for 75-85% of total variance in the two sources. Roemer (2002) compares earnings measures in both the SIPP and March CPS to those from social security earnings records. His general findings for the CPS seem quite consistent with those in Abraham, Haltiwanger, Sandusky, and Spletzer (2009): that workers with marginal labor market attachment are likely to under-report formal jobs, and that jobs in occupations in which non-employee status or off-the-books wages are more likely are more likely to be reported in the CPS than in the administrative data. He finds smaller discrepancies in SIPP than in the CPS.

3 Data

We focus on comparing earnings measures from UI data to those drawn from the SIPP. We plan to use the 1996, 2001 and 2004 SIPP panels, but in this preliminary draft we present results using only the 1996 panel. For now, we also restrict our sample to men aged 25-59. We use data from the Longitudinal Employer-Household Dynamics (LEHD) database, which draws much of its data from complete sets of UI earnings records for partner states. Currently 48 of the 50 U.S. states (along with DC, Puerto Rico, and the Virgin Islands) have joined the partnership, but the version of the database that we have access to includes records for 31 states. The database includes earnings for years 1990 to 2004 when available, but fewer states are included in earlier years, so here we use UI earnings from the 18 states for which we have complete earnings data for all four quar-

ters of years 1996-1999.[2] While the core of the LEHD data base comes from UI records on workers' earnings, these records have been matched to other characteristics of employers gathered in quarterly administrative UI reports and through Census Bureau business censuses and surveys. The database also incorporates basic demographic data on workers that are drawn from a combination of Census surveys and other sources of administrative records.

We would expect UI and SIPP earnings measures to match only where they cover the same set of jobs, so it is important to understand where coverage differs. Earnings in SIPP are collected for three different types of employment—wage and salary employment, non-farm self-employment, and farm self-employment. We use only earnings from wage and salary employment because that most closely matches UI coverage. We plan to examine whether earnings from self-employment in an incorporated business can account for some of the men who have employment in the UI data but not in SIPP, but have not done so in this draft.

State UI programs cover over 96% of wage and salary employment and have fairly comparable coverage from state to state (Bureau of Labor Statistics 1997). Certain types of jobs and workers have lower coverage rates and their coverage varies more from state to state. These include employees of small agricultural employers, student employees, insurance and real estate agents who work on commission, employees of nonprofit organizations (particularly those operated primarily for religious purposes), officers of corporations, elected officials, and service as a member of a state National Guard.[3] Federal employees and ex-military members are covered by separate federal UI programs, and data from these programs are not currently included in the LEHD database. In addition, work as an independent contractor and self-employment are not covered by state

[2]We are working on incorporating data from a new snapshot of the database that includes earnings for 1990-2008 for 47 states. This expands the set of states with complete data for 1996-1999 to 28.

[3]U.S. Department of Labor (2007)

UI programs.

Each SIPP panel comes from a nationally representative sample of households in the civilian non-institutionalized U.S. population. Sample members within each panel are randomly divided into four rotation groups of roughly equal size. One rotation group is interviewed each month, and asked to report information about the previous four months. Thus, each sample member is interviewed once every 4 months over the life of the panel, which for the 1996 panel was 48 months. Individuals who change residence or who move out of a sample household are generally followed and interviewed at the new address if possible.

One of the advantages of using SIPP is that earnings data are collected for specific jobs with a monthly reporting period, allowing close comparison of jobs and earnings reported in the two sources. SIPP's monthly reference period for earnings make it simple to aggregate up to quarters in SIPP to match the UI reporting periods. Our basic strategy is to construct comparable quarterly and annual measures of earnings and jobs held from the SIPP and UI data. While in the end our interest is more in year-to-year changes in annual earnings, making the comparison first on a quarter-by-quarter basis gives us some leverage to identify the sources of differences in annual earnings.

One drawback to the way in which the SIPP data are collected is that the different starting and ending dates of the rotation groups mean that full calendar year measures are not available for part of the sample at the beginning and end of the panel. So we lose half of the sample (rotation groups 3 and 4) for calendar year 1996 because members of those rotation groups were not asked to provide data for the first month or two of that year. Similarly, we lose one quarter of the sample (rotation group 1) for 1999 because that rotation group was not asked to provide data for December of 1999. For this reason, in much of what follows we concentrate on estimates based on 1997 and 1998 data.

Roughly 15% of the 1996 SIPP sample cannot be linked because respondents did not

provide the information required for Census to link the data. We refer to the part of the sample that did provide matching information as those who are potentially matched. We estimate logistic models of the propensity to provide matching information among residents of our 18 sample states to try to adjust for this first stage of selection into our matched sample. The variables used in the propensity score model include age and its square, whether the sample member was interviewed directly or by proxy, share of quarters worked in first year in SIPP, and state of residence. We use the SIPP calendar-year survey weights in presenting basic sample statistics to illustrate differences between the overall and matched samples (Table 1), but the other tables use weights that have been adjusted based on our estimates of matching propensity.

In practice only a subset of individuals with the information required for matching are found in the UI earnings data. There are several possible reasons for this. One is that we currently have UI earnings data for a limited number of states, and so some of those who do not appear in the UI data may work in other states. To eliminate the bulk of such non-matches, we restrict our comparison to SIPP respondents living in one of the states for which we have UI data. The first panel of Table 1 gives overall sample sizes for our two data sources, and then for the subsets we use to do the analysis.

A second reason for a non-match to the UI data is that the person is not working in a UI-covered job, or is not working at all. While in the SIPP we can distinguish between periods in which an individual reports no work and periods in which data are unavailable for that person, the distinction between missing and zero earnings is not as clean in the UI data. To try to identify where differences are caused by incomplete UI coverage, we use SIPP variables that classify jobs by type of employer, industry, and occupation to try to identify which sample members are likely to be affected.

We are interested in making comparisons of earnings at the job-level, as well as total earnings summed across jobs. Because job identifiers are not common across the two

data sources, we need some basis for deciding which jobs to compare for those holding more than one within a time period. For this purpose, we rank jobs according to earnings within the time period and then restrict the comparison to jobs that account for at least 70 percent of total earnings, which we will refer to as "main jobs." We plan to use industry and employment history to examine whether in fact the main job in both sources appears to match but have only made preliminary comparisons of industry at this point.

4 Preliminary results

We first provide some descriptive statistics for alternative samples. We then turn to a more careful examination of differences in earnings levels and earnings changes. Finally we compare instability estimates from the different samples and try to identify the sources of those differences.

4.1 Sample characteristics

Table 1 presents descriptive statistics that are meant to illustrate basic differences between the overall 1996 SIPP panel sample, the overall UI earnings sample, and the part of those samples that we can match. The first two columns use measures constructed from SIPP data, while the last column is based on UI earnings from the LEHD data base. As the first panel illustrates, just over half of the SIPP sample lives in one of the states for which we have UI data. While these are unweighted counts, the weighted share is essentially the same.

The second and third panels give mean age and employment rates for the various cuts of the data. There is little difference in the mean for either variable between the overall SIPP sample and the portion of the sample that provided matching information.

The portion of the SIPP sample that matches to the UI records is slightly younger than the overall sample, but has a substantially higher employment rate (where anyone with positive earnings for the year is counted as employed). The higher employment rate is to be expected given that the UI records include only those with covered employment. Note that average age is the same for each of the cuts of the data that require UI employment (both UI measures and the SIPP measure for the matched sample), so higher age in the SIPP sample appears to simply reflect slightly lower employment rates among older men. The fact that 6% of SIPP sample members who match to the UI data do not report any wage and salary employment in the SIPP points to differences across the two sources in who is identified as employed.

The other panels in Table 1 present statistics that condition on having positive earnings for the year. Among those employed, more jobs are reported in the UI data than in the SIPP data. In both data sources the majority of men have only a single job, but multiple jobs are more common and the average number of jobs held by multiple-job holders is greater in the UI data. The potentially matched sample has slightly higher earnings than the overall SIPP sample, but the difference is only about two percent. For men in this age range and living in our LEHD states, mean UI earnings are about seven percent higher than mean SIPP earnings. But where the statistics are based on the same set of people, the difference is almost 10 percent.

Earnings growth is where aggregate statistics for the two samples show the largest differences. Log earnings growth as measured in the SIPP is higher in our LEHD states than it is in the rest of the sample, but within these states growth is similar for the overall and potentially matched samples. The UI measure of earnings growth is lower than that based on SIPP data both overall and particularly for the sample that matches. We are not sure why this is, though we do note that mean earnings growth varies substantially from year to year in both sources.

Aside from substantial differences in wage growth, these aggregate statistics suggest modest differences between the two data sources. We now turn to making comparisons at the individual level where large differences are fairly common.

4.2 Consistency of earnings and jobs across data sources

We look first at the number of jobs reported in the two sources. We use the potentially matched sample in these calculations and treat anyone who could have matched to the UI data but did not as not having a job. We exclude those who report working for the federal government on their highest earnings job, as federal workers are clearly not covered by the UI data we have. [4] When we turn to earnings comparisons, we additionally restrict comparisons to those with positive non-imputed earnings from both sources.

Table 2 gives the joint distribution of the number of jobs reported at both the quarterly and annual level. We group those with earnings from two or more jobs together, in part because we have earnings from at most two jobs in a quarter for SIPP respondents for whom the quarter is covered by a single reference period. In more detailed job counts, the two sources agree more often on whether someone was a multiple-job holder than on how many jobs they held. Most sample members have a single job according to both sources. During a quarter, 65% of potentially matched sample members have exactly one job in the UI data, and 71% have a single SIPP job. 56% have one job in both data sources. Unsurprisingly, having a single job over an entire year is somewhat less common, but is still the modal case in both data sources. Having no job and having more than one job are both more common in the UI data, though the difference in multiple job holding is minor. The most common type of disagreement between sources is where those with one job in the SIPP do not show up in the UI data.

[4] Less than one quarter of this group matches to the UI data, and those that do match have dramatically lower earnings in the UI data than in SIPP.

An individual can report two jobs in a year either because (i) they worked for a single employer at any one time, but changed employers at some point; or (ii) they worked for two employers at one time. In this first case, we would find earnings from two employers only in one quarter of the year, while in the latter case there could be as many as four quarters with multiple jobs. If SIPP respondents more accurately report changes in employer than they do jobs held simultaneously, we would expect a greater discrepancy in the quarterly measures than in the annual measures. There is some indication of such a pattern: in the quarterly measures, only 40% of those who are multiple-job holders in the UI data also have more than one job in the SIPP, while in the annual measures the rate of agreement is 57%. Conditioning on SIPP multiple-job holding similarly shows higher rates of agreement in the annual measures. We intend to look at this more carefully using information from the two sources on when jobs start and stop.

We now turn to differences in earnings in the two sources. In doing so, we restrict our sample to individuals who have non-imputed earnings from both sources (and live in one of our LEHD states). Table 3 categorizes the difference in log earnings. The gap between the two sources is greater than 10% for about 3/5ths of the sample. In all sectors, UI earnings are much more likely to exceed SIPP earnings than the converse.

We examine differences across sectors as a way of generating hypotheses about what might generate the large inconsistencies in earnings. Certain sectors stand out as having particularly low rates of agreement: finance, insurance, and real estate (FIRE); construction; and personal services. These same sectors also rank highest based on the share of cases with SIPP earnings more than 10% above UI earnings. As mentioned earlier, insurance and real estate agents who work on commission have relatively low rates of UI coverage, which may account for FIRE's relatively high share of cases with SIPP earnings exceeding UI earnings. Other factors seem more likely to be relevant for construction and personal services: for example, personal services includes several industries in

which tip income may be important (hotels, parking lots, beauty shops), while in construction we might expect to see a fair amount of either off-the-books jobs or jobs in which work is done as an independent contractor. Educational services is the one sector in which the majority of sample members have close agreement on earnings from the two sources, perhaps reflecting the very standardized and predictable salary structure in teaching. (It is also the sector in which the two sources are most likely to agree on main-job industry.)

The two manufacturing categories and business services have the highest share with substantially greater UI earnings. Employees in manufacturing are less likely than those in other sectors to have more than one job (according to both data sources), so this seems likely to be a result of differences in earnings measured on their manufacturing job. One explanation might be underreporting of overtime income in SIPP. Business services includes industries such as advertising, personnel supply services, and computer services. It is not clear to us what might make that sector have particularly high UI earnings relative to SIPP earnings.

We turn to regressions to look more systematically for factors explaining the differences we observe. Table 4 presents selected coefficients from regressions using the difference in log earnings from the two sources as the dependent variable. The regressions include year, state, industry, and occupation controls in addition to the variables listed in Table 4. The industry and occupation coefficients are presented in Tables 5 and 6 respectively. The dependent variable is the log of SIPP earnings minus the log of UI earnings. UI earnings are higher on average than SIPP earnings, so the mean of the dependent variable is negative. Positive coefficients indicate characteristics that are associated with relatively high earnings in SIPP, and thus on average a narrower gap.

The first column gives results using quarterly data while the second uses annual data. The standard errors account for repeated observations on the same person. The

quarterly results are based on a somewhat larger sample because the rotation groups with reference periods that do not cover all of the months in calendar years 1996 and 1999 still provide 3 quarters of data for those years. Job characteristics such as industry, occupation, and class of worker are based on the highest-earning job for the time period.

The coefficients on personal characteristics such as age, education, and whether foreign-born are generally small and in most cases insignificant. Men with more continuous employment—measured as the share of quarters worked during the period that they are observed in the SIPP data—have higher SIPP earnings relative to their UI earnings in the annual regression, but that effect is not found in the quarterly data. Everyone in the quarterly regression is employed according to both sources in that quarter, so the lack of an effect in the quarterly estimates indicates that in quarters with work, the gap does not differ systematically by how continuously people work. The annual regressions include some men with zero SIPP quarters, so in the absence of any quarterly effect, the positive coefficient in the annual regression reflects positive UI earnings in some of the quarters with zero SIPP earnings.

The gap between UI and SIPP earnings is somewhat larger in the fourth quarter than in other quarters. One explanation might be under-reporting of seasonal fluctuations in pay in the SIPP—for example, end of year bonuses. An alternative explanation might be under-reporting of short-term seasonal jobs at this time of year—for example, short-term retail jobs over the holiday season. We plan to investigate this further, though the effect is fairly modest for the average sample member.

The variables with significant explanatory power in these regressions are generally those associated with the number and types of jobs held. The coefficient for state and local government employees suggests that they have higher UI earnings relative to SIPP earnings than private sector workers (the omitted group) but this is while controlling for industry and occupation. Most state and local government (SLG) employees are in

the industries education services (about 35% of all SLG employees) or public administration (about 40%), and they account for over 75% of employment in both of those industries once federal workers are excluded. Looking at the combined effects of industry and class of worker, the industry coefficients in Table 5 more than offset the class of worker coefficient for both of these groups. Looking at the other industry coefficients, the positive significant effects for construction, FIRE, and personal services are consistent with the patterns we saw in Table 3. None of the occupation coefficients in Table 6 are significantly different from zero.

The coefficients on the variables describing multiple job holders are all quite large and significant. Overall they suggest that earnings on secondary jobs that appear in one data source but not the other are an important source of discrepancies. We use predictions based on the annual regression results to illustrate this in Figure 2. The bottom three bars illustrate that, holding the other characteristics in the regression constant, average differences in earnings for those reporting the same number of jobs in both sources on average have relatively small discrepanacies in earnings. The larger bars on the SIPP>UI side of the graph suggest that the jobs missing in the UI data tend to be higher earnings jobs than the jobs missing in the SIPP data.[5]

We have run the same set of regressions using the difference in earnings on a worker's main job, where we include an observation only if in both data sources the highest earnings job accounts for at least 70% of earnings and has earnings exceeding a cut-off value. Our interest is in narrowing down the sample to observations for which the main job

[5]We have also estimated models in which we allow for interactions between the UI and SIPP measures of multiple-job holding to relax the implicit assumption that the change in the gap associated with an additional job in one data source does not depend on the number of jobs in the other data source. The results of this messier model indicate that the simpler model is a reasonable summary of the relationship. That is, where the job count categories match, the average gap is slightly smaller when both sources have earnings from 2+ than when there both have a single job. If there are more SIPP jobs than UI jobs, SIPP minus UI earnings are much more positive than when the job counts agree. If there are more UI jobs, the gap is wider. But having more reported SIPP jobs than UI jobs has a much bigger effect on the gap than the converse.

from the two data sources will in most cases be the same. Eliminating observations with very low earnings on the highest paying job should get rid of some of the observations for which the job differs because a high-earnings job does not appear in one of the sources. We experiment with two values: at least $5,000 in annual earnings ($1,250 quarterly) and at least $20,000 in annual earnings ($5,000 quarterly). The $20,000 cut-off seems more reliable from this perspective, but also eliminates workers who truly do not have a high-earnings job.

Most of the coefficients in these regressions are quite similar to those in Table 4, so in Table 7 we report only the coefficients on the number of jobs held, because their interpretation changes. Since the dependent variable here is the difference in earnings on a single job, differences in which jobs appear in the two data sources affect the results only if differences in coverage mean that the main jobs identified in the two sources are not in fact the same job. If we do in fact have earnings from the same job, then differences associated with multiple job holding either represent reporting effects—e.g. the additional burden of reporting on more than one job affects the accuracy of reports on all jobs—or something about the kinds of main jobs held by those with other jobs—e.g. they are jobs with irregular earnings that are more difficult to report accurately. One form of reporting error in SIPP might be for respondents to simplify their task by summing up earnings from several short term jobs and reporting them as one. This would appear as more jobs in the UI data but higher earnings on the main job in SIPP reports.

The annual estimates are consistent and more substantial in both panels, so we concentrate on those. In both panels, multiple-job holding recorded in the UI data (holding the number of SIPP jobs reported constant) is associated with lower measured earnings on the main UI job, or on average, a narrower gap between sources. This is in contrast to its association with a wider gap in total earnings. Similarly, multiple-job holding in the SIPP is associated with lower earnings on the main SIPP job which would on average

widen the gap.

4.3 Comparing measures of earnings instability

We now examine how the differences in earnings levels we have noted across sources affect estimates of earnings instability. Economists have used a variety of approaches to estimate the variance of transitory shocks to earnings. Several papers by Gottschalk and Moffitt (e.g. Moffitt and Gottschalk (2002), Moffitt and Gottschalk (2008)) have focused on using the long panel of earnings provided by the PSID to carefully distinguish transitory from more permanent changes. Other authors (e.g. Cameron and Tracy (1998), Shin and Solon (2008)) have used simpler measures that can be applied to short panels, which is the approach we use here. We follow the methodology in Shin and Solon (2008) to construct estimates of earnings instability. Essentially we are measuring the cross-sectional spread in changes in earnings, where increases in the instability of earnings would result in an increase in this spread. To do this, we first regress the annual change in log of wage and salary earnings on a quadratic function in age, which removes year-to-year and life-cycle variation in mean earnings changes. We then use the standard deviation of the residuals from this regression as our measure of earnings instability.

We measure age as of the last month in each calendar year and in computing instability measures use only individuals who are 25-59 in both years. We exclude individuals with zero wage and salary earnings and then trim the top and bottom 1 percent among those with positive earnings in each year. We also exclude individuals with zero calendar year weights within the two-year period. After those exclusions, we have a sample of about 4,000 observations per year for the matched sample. Excluding records with imputed earnings for any of the months within the two-year period needed reduces our sample sizes to about 2,500 observations per year. When using the matched sample to produce UI-based measures of instability the sample sizes are essentially the same, but

for the one estimate for all men 25-59 in the LEHD database (row 1 of Table 8) our sample size is about 24 million observations.

Table 8 gives several alternative instability estimates. As in Table 1, the first line presents results using the overall sample of men while the second line gives estimates for all SIPP sample members for whom matching information is available. Any selection in providing matching information appears to have little effect on instability estimates, though restricting the sample to the LEHD subset of states does raise instability slightly. The third row uses only SIPP sample members that also appear in the UI data. This group has slightly lower SIPP income instability than the potentially matched sample, and substantially lower instability than the overall LEHD set of workers. The first panel uses all of the available data for a given sample, while the second panel excludes observations for which earnings were imputed in some months of the year. The lower estimates of earnings instability in the second panel indicate that earnings imputation in the SIPP tends to lead to more variable year-to-year changes in earnings.

Even using a consistent set of men in the matched sample, we find much lower instability using earnings as measured in the SIPP than we do using UI earnings measures. While we use the same basic methodology in producing estimates from the two data sources, when we trim the top and bottom 1% of earnings we end up trimming at different levels of the earnings distribution because the UI data have more very low and very high values than the SIPP data do. To examine how important this is, we compute instability using the first and 99th percentiles from the SIPP distribution to trim the UI earnings distribution. This eliminates just over half of the difference in instability levels. Reducing the level at which we trim the top part of the distribution has quite modest effects on instability estimates. Changing only the trim level at the bottom of the distribution gives essentially the same level of instability as changing both trim levels. Thus the reduction comes from eliminating people who have relatively low UI earnings in at

least one of the years used in the change.

As a first attempt to guage the importance of differing job reports in generating differences in instability, we further restrict our sample to those reporting one job in both sources and in both years of the difference. This substantially narrows the gap between instability measures, but does not eliminate it.

Based on these exercises, we are now investigating the characteristics of the lower part of the earnings distribution in the two data sources. We are currently focusing on differences in earnings changes and in the consistency of job reports across sources among those who appear in the lower part of the UI earnings distribution.

5 Preliminary Conclusions

We find that in levels, administrative earnings from UI records are higher on average than earnings as measured in SIPP, consistent with findings from other comparisons of survey and administrative earnings measures. While UI earnings exceed SIPP earnings by more than 10% for almost half of the matched sample, for a different 20% of the sample SIPP earnings are substantially higher than UI earnings. The average number of jobs reported in the two sources do not differ much, but at a micro level the counts are often not consistent. Differences in the number of jobs reported in the two sources are associated with substantial differences in earnings.

Clearly some of the differences arise from differential coverage of particular types of jobs, and the fact that we miss jobs in neighboring states if they are not in our LEHD sample. We are currently incorporating data for a more complete set of states which will allow us to measure how much we miss by looking at a subset of states. We also plan to look more closely at those who report construction employment, as this is an industry with a relatively wide gap in measures from the two sources, and also one

in which work arrangements seem more likely to vary between work as an employee, as an independent contractor, or doing smaller jobs off the books. We also intend to investigate our finding that differences between the two sources are larger in the fourth quarter of the year because this may give clues to what types of jobs and earnings are more likely to be reported in one source than the other.

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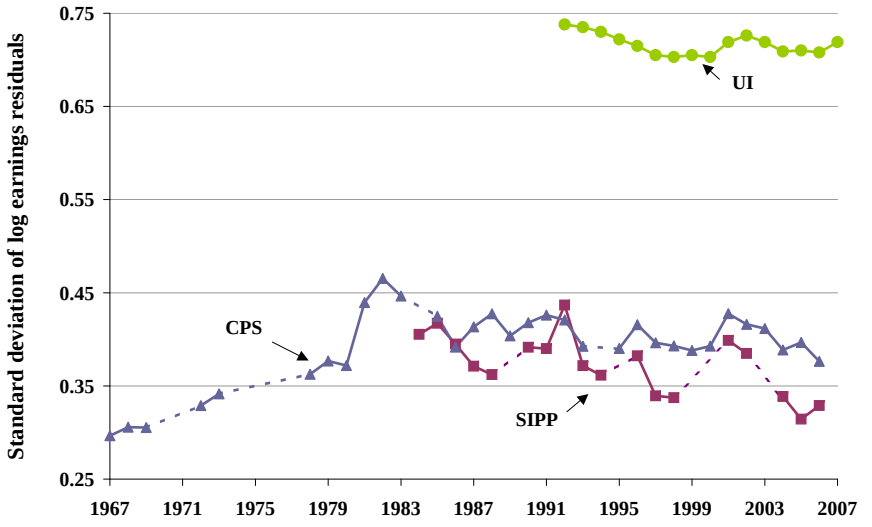
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Figure 1: Instability estimates using CPS, SIPP, and UI data



Notes: The figure plots the standard deviation of residuals from a regression of the change in log earnings on a quadratic in age. Dashed segments of the line represent interpolation for years in which estimates are not available.

Figure 2: Average earnings differences by number of jobs reported

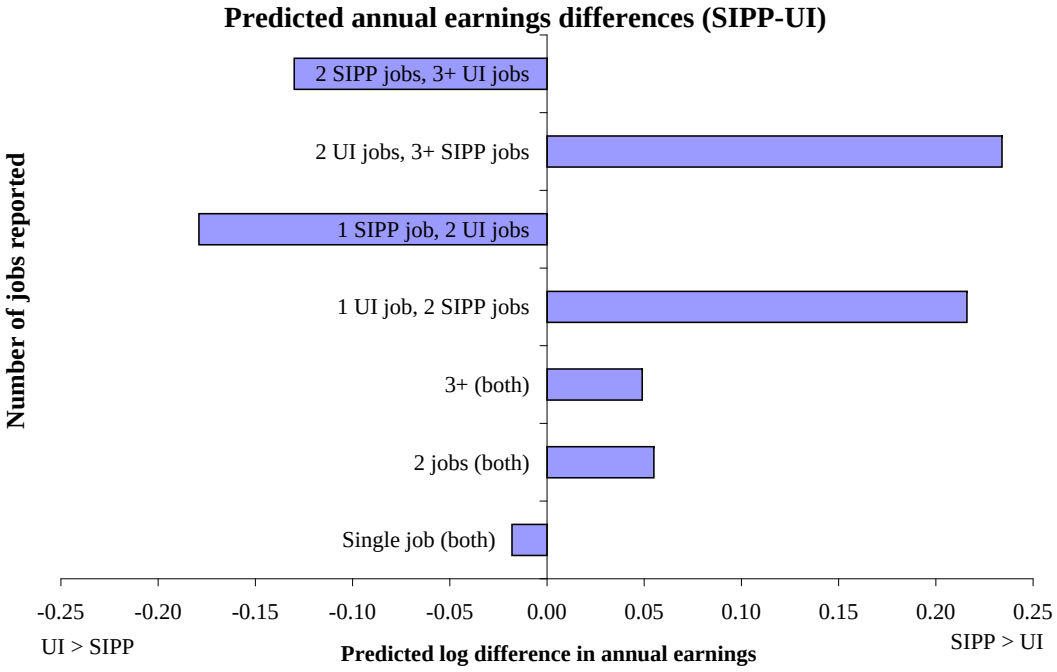


Table 1: Sample statistics

	SIPP Measures		UI Measures
	All states	**LEHD states**	
1. Sample sizes			
All	20,342	10,800	27,985,202
Potentially matched	17,016	8,903	
Matched		6,852	6,852
3. Share employed during year			
All	0.82	0.82	
Potentially matched	0.82	0.83	
Matched		0.94	1.00
6. Annual earnings (excluding 0s)			
All	35,573	35,755	38,193
Potentially matched	36,281	36,585	
Matched		36,779	40,271

Table 2: Joint distribution of jobs held per time period in SIPP and UI data

	Number of jobs in UI data		
Number of jobs reported in SIPP	**None**	**Single job**	**2 or more**
Quarterly measure			
None	0.17	0.04	0.01
1	0.08	0.58	0.05
2 or more	0.01	0.03	0.04
Annual measure			
None	0.13	0.04	0.01
1	0.06	0.46	0.09
2 or more	0.01	0.06	0.13

Notes: Includes SIPP sample members who provided information needed to match to the UI data and lived in one of the LEHD states used. Excludes those who reported working for the federal government on their highest-earnings job. None for UI jobs indicates no match to UI earnings data in that quarter or year.

Table 3: Distribution of difference between SIPP and UI annual earnings, by sector

Industry category	**UI > 10% more**	**Within 10%**	**SIPP > 10% more**
Construction	0.39	0.37	0.25
Nondurable manufacturing	0.45	0.43	0.12
Durable manufacturing	0.48	0.42	0.10
Transportation, communication, utililties	0.45	0.40	0.15
Wholesale trade	0.44	0.42	0.14
Retail trade	0.41	0.39	0.20
Finance, insurance, and real estate	0.41	0.35	0.24
Business services	0.47	0.38	0.15
Personal services	0.38	0.37	0.24
Health services	0.46	0.41	0.13
Professional services	0.38	0.46	0.16
Education services	0.33	0.51	0.16
Public administration	0.44	0.43	0.13
All	0.43	0.41	0.16

Notes: Comparisons are based on log differences in earnings, with a log difference of 0.1 labeled as a 10% difference. Workers are assigned to industry categories based on the industry reported in SIPP for their highest earning job. Includes matched sample members who were living in one of the LEHD states used and had positive earnings in both SIPP and UI reports for calendar year. Those reporting they worked for the federal government are excluded from the calculations.

Table 4: Log of SIPP earnings minus log of UI earnings (selected coefficients)

	Quarterly	Annual
SIPP data from proxy interview	-0.022**	0.001
	(0.007)	(0.014)
SIPP data from >1 interview	-0.033***	
	(0.004)	
Share of quarters worked 1996-1999 (SIPP)	0.001	0.180**
	(0.040)	(0.068)
Age	-0.002***	-0.002*
	(0.000)	(0.001)
Foreign-born	-0.038*	-0.066*
	(0.016)	(0.028)
High school graduate	-0.007	-0.007
	(0.016)	(0.027)
Some college but no bachelor's degree	0.014	0.015
	(0.016)	(0.026)
College graduate	0.043*	0.034
	(0.019)	(0.032)
Advanced degree holder	0.036	0.039
	(0.021)	(0.037)
2nd quarter	-0.006	
	(0.006)	
3rd quarter	-0.002	
	(0.006)	
4th quarter	-0.045***	
	(0.006)	
State and local government worker	-0.060**	-0.164***
	(0.023)	(0.050)
Two jobs reported in SIPP (2+ for quarterly reg)	0.306***	0.234***
	(0.019)	(0.024)
Three or more jobs reported in SIPP		0.413***
		(0.042)
Two jobs in UI records (2+ for quarterly reg)	-0.227***	-0.161***
	(0.014)	(0.020)
Three or more jobs in UI records		-0.346***
		(0.037)
N	60,305	12,103

Notes: Uses 1996-1999 earnings for matched sample members living in one of the LEHD states used, and having positive non-imputed earnings in both SIPP and UI data in that time period. State/local government dummy is based on class of worker in highest-earnings job. Additional controls include dummies for year, SIPP state of residence, and SIPP industry and occupation on highest-earnings job. Standard errors account for repeated observations on same person.

Table 5: Industry effects from log earnings difference regressions

	Quarterly	Annual
Construction	0.081***	0.106**
	(0.018)	(0.034)
Nondurable manufacturing	-0.014	0.011
	(0.017)	(0.031)
Durable manufacturing	-0.019	-0.021
	(0.015)	(0.026)
Transportation, communications, and utilities	0.001	0.011
	(0.016)	(0.028)
Retail trade	0.032	0.058
	(0.020)	(0.033)
Finance, insurance, and real estate	0.056*	0.093*
	(0.026)	(0.043)
Business services	0.024	0.015
	(0.019)	(0.034)
Personal services	0.046*	0.099*
	(0.021)	(0.039)
Health services	-0.032	-0.034
	(0.025)	(0.039)
Professional services	-0.001	-0.005
	(0.020)	(0.038)
Education services	0.095**	0.225**
	(0.033)	(0.075)
Public administration	0.078**	0.249***
	(0.030)	(0.062)

Notes: Based on same regressions as Tables 4 and 6. Based on 1996-1999 earnings for matched sample members living in one of the LEHD states used, and having positive earnings in both SIPP and UI data in that calendar year. Coefficients for agriculture, mining, and social services/ membership organizations are not reported because of small cell sizes. Standard error estimates account for repeated observations on same person.

Table 6: Occupation effects from log earnings difference regressions

	Quarterly	Annual
Engineering, math and scientific professional occupations	-0.011	-0.041
	(0.014)	(0.026)
Teachers	-0.009	-0.028
	(0.028)	(0.063)
Technicians	-0.013	-0.027
	(0.017)	(0.029)
Sales occupations	0.001	-0.017
	(0.019)	(0.033)
Administrative support	0.026	-0.053
	(0.015)	(0.031)
Protective services	-0.030	-0.090
	(0.023)	(0.048)
Other service occupations	0.021	-0.000
	(0.022)	(0.038)
Mechanical and repair	-0.015	-0.032
	(0.013)	(0.024)
Construction trades	0.029	0.059
	(0.019)	(0.039)
Precision production	-0.004	-0.028
	(0.016)	(0.033)
Machine operators	-0.011	-0.048
	(0.022)	(0.039)
Other production occupations	0.007	-0.012
	(0.019)	(0.041)
Transportation occupations	0.020	-0.012
	(0.017)	(0.031)
Laborers	0.001	0.029
	(0.019)	(0.035)

Notes: Based on same regressions as Tables 4 and 5. Based on 1996-1999 earnings for matched sample members living in one of the LEHD states used, and having positive earnings in both SIPP and UI data in that calendar year. Coefficients for the following occupations are not reported because of small cell sizes: health occupations, private household, extractive, lawyers, other professional specialty, writers/artists/entertainers, and farm/agricultural. Standard error estimates account for repeated observations on same person.

Table 7: Log of SIPP earnings minus log of UI earnings on main job

	Quarterly	Annual
Using $5,000 annual cutoff:		
Two jobs reported in SIPP (2+ for quarterly reg)	0.033*	-0.085***
	(0.013)	(0.016)
Three or more jobs reported in SIPP		-0.087*
		(0.042)
Two jobs in UI records (2+ for quarterly reg)	0.011	0.077***
	(0.010)	(0.015)
Three or more jobs in UI records		0.093***
		(0.026)
Using $20,000 annual cutoff:		
Two jobs reported in SIPP (2+ for quarterly reg)	0.002	-0.085***
	(0.009)	(0.014)
Three or more jobs reported in SIPP		-0.121***
		(0.034)
Two jobs in UI records (2+ for quarterly reg)	0.020*	0.077***
	(0.009)	(0.012)
Three or more jobs in UI records		0.082***
		(0.024)

Notes: Based on 1996-1999 main-job earnings for matched sample members living in one of the LEHD states used. Includes only observations for which, in both data sources, the highest-earnings job accounts for at least 70% of total earnings and has earnings above the given threshold on an annual basis. Excludes imputed values. Regressions include same set of regressors as those in Tables 4 - 6.

Table 8: Instability estimates

	SIPP		LEHD
	All states	**LEHD states**	
Using source-specific trim levels			
All	0.381	0.391	0.684
Potentially matched	0.383	0.393	
Matched		0.365	0.550
Using SIPP trim levels for both SIPP and LEHD estimates			
Matched		0.365	0.448
Single job in both sources, both years			
Matched		0.268	0.316

Notes: Figures represent the standard deviation of residual log earning changes for 1997-1998. Observations in the top and bottom tails of the earnings levels distributions are excluded ("trimmed") based on the first and 99th percentiles of the SIPP and LEHD earnings distributions as indicated in panel headings. Samples exclude observations with imputed SIPP data, except in the case of the overall LEHD sample.